Universidad de Costa Rica

PI de redes y sistemas operativos

CI0123

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Semestre:

I-2026

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Using configuration from /home/brandondelg/easy-rsa/pki/easy-rsa-12134.XlAPJh/tmp.GLKr3S
Enter pass phrase for /home/brandondelg/easy-rsa/pki/private/ca.key:
407741C283750000:error:0700006C:configuration file routines:NCONF_get_string:no value:../crypto/conf/conf_lib.c:315:group=<NULL> name=unique_subject
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'redes'
Certificate is to be certified until Jul 18 06:06:22 2028 GMT (825 days)

Write out database with 1 new entries
Data Base Updated

brandondelg@pop-os:~/easy-rsa$ ./easyrsa build-client-full cliente nopass
Using SSL: openssl OpenSSL 3.0.2 15 Mar 2022 (Library: OpenSSL 3.0.2 15 Mar 2022)
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Using configuration from /home/brandondelg/easy-rsa/pki/easy-rsa-12262.1f4riW/tmp.othjls
Enter pass phrase for /home/brandondelg/easy-rsa/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'cliente'
Certificate is to be certified until Jul 18 06:08:28 2028 GMT (825 days)

Write out database with 1 new entries
Data Base Updated

brandondelg@pop-os:~/easy-rsa$
```

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Setting up easy-rsa (3.0.8-1ubuntu1) ...
Setting up opensc (0.22.0-1ubuntu2.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for libc-bin (2.35-0ubuntu3.10) ...
brandondelg@pop-os:~$ ls /usr/share/easy-rsa/
easyrsa  openssl-easyrsa.cnf  vars.example  x509-types
brandondelg@pop-os:~$ make-cadir ~/easy-rsa
brandondelg@pop-os:~$ cd ~/easy-rsa
brandondelg@pop-os:~/easy-rsa$ ls
easyrsa  openssl-easyrsa.cnf  vars  x509-types
brandondelg@pop-os:~/easy-rsa$ ./easyrsa init-pki

init-pki complete; you may now create a CA or requests.
Your newly created PKI dir is: /home/brandondelg/easy-rsa/pki


brandondelg@pop-os:~/easy-rsa$ ./easyrsa build-ca
Using SSL: openssl OpenSSL 3.0.2 15 Mar 2022 (Library: OpenSSL 3.0.2 15 Mar 2022)

Enter New CA Key Passphrase:
Re-Enter New CA Key Passphrase:

Easy-RSA error:

Passphrases do not match.

brandondelg@pop-os:~/easy-rsa$ ./easyrsa build-ca
Using SSL: openssl OpenSSL 3.0.2 15 Mar 2022 (Library: OpenSSL 3.0.2 15 Mar 2022)

Enter New CA Key Passphrase:
Re-Enter New CA Key Passphrase:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
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Common Name (eg: your user, host, or server name) [Easy-RSA CA]:CI0123 CA

CA creation complete and you may now import and sign cert requests.
Your new CA certificate file for publishing is at:
/home/brandondelg/easy-rsa/pki/ca.crt
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Pruebas del pki

Para instalar apache:

sudo apt update

sudo apt install apache2 –y

Copiar los certificados a apache

sudo cp pki/issued/redes.crt /etc/ssl/certs/

sudo cp pki/private/redes.key /etc/ssl/private/

sudo cp pki/ca.crt /etc/ssl/certs/

Activar ssl en apache

sudo a2enmod ssl

Activar el sitio SSL

sudo a2ensite default-ssl

Configurar el certificado en Apache
sudo nano /etc/apache2/sites-available/default-ssl.conf

Y dentro del archivo agregar debjao de SSLEngine on:

SSLCertificateFile /etc/ssl/certs/redes.crt

SSLCertificateKeyFile /etc/ssl/private/redes.key

SSLCACertificateFile /etc/ssl/certs/ca.crt

Elimiar el certificado por defecto

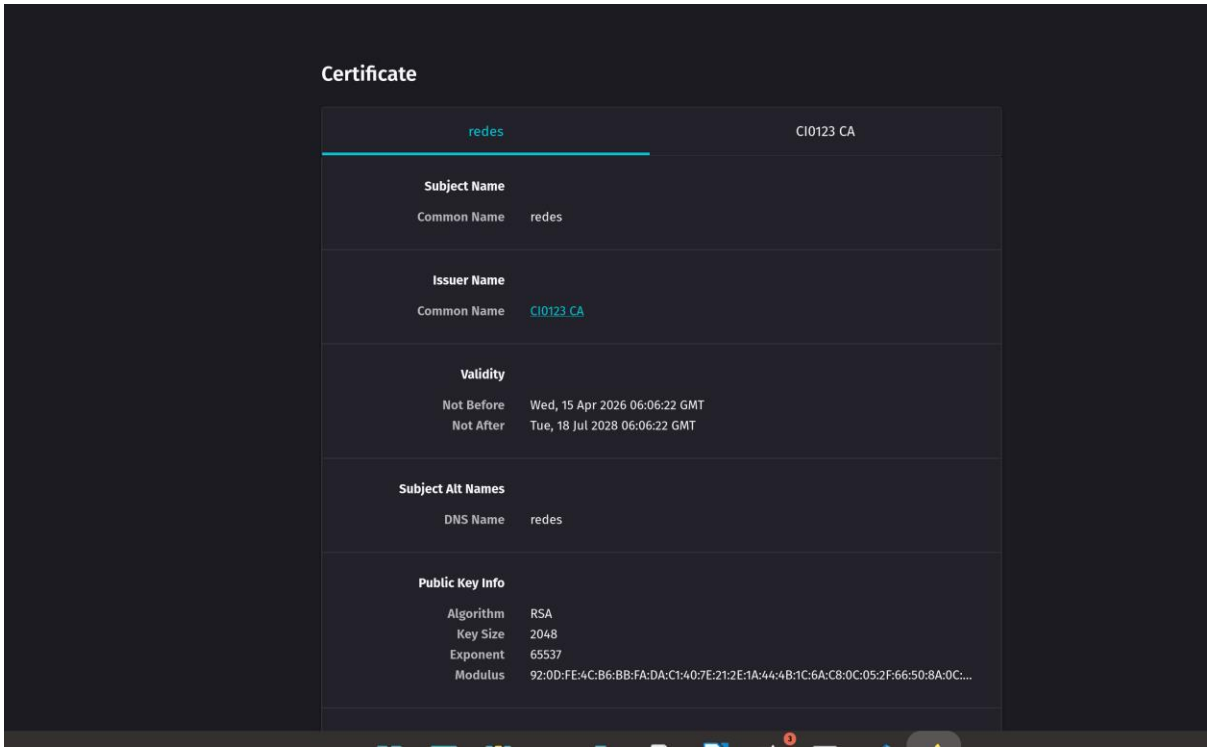
Reiniciar el servidor

sudo systemctl restart apache2

Probar en el navegador:

https://localhost

Pruebas:


Certificate
redes
CI0123 CA
Subject Name
Common Name redes
Issuer Name
Common Name CI0123 CA
Validity
Not Before Wed, 15 Apr 2026 06:06:22 GMT
Not After Tue, 18 Jul 2028 06:06:22 GMT
Subject Alt Names
DNS Name redes
Public Key Info
Algorithm RSA
Key Size 2048
Exponent 65537
Modulus 92:0D:FE:4C:B6:BB:FA:DA:C1:40:7E:21:2E:1A:44:4B:1C:6A:C8:0C:05:2F:66:50:8A:0C:...